

An Empirical Geospatial Framework for Estimating Serviceable Satellite Broadband Subscribers in Terrestrial Coverage Gaps: A Case Study of Remote Areas in Thailand

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Abstract

Despite high aggregate broadband achievement, many emerging economies retain residual terrestrial coverage gaps in mountainous, forested, border and island areas where conventional network expansion is economically unviable. Low Earth orbit (LEO) satellite broadband has emerged as a credible alternative, yet the magnitude and spatial distribution of potentially serviceable subscribers in such gaps remain empirically underspecified. Using remote areas of Thailand as a case study, this paper proposes an empirical geospatial framework that integrates building footprints from Google Open Buildings, terrestrial mobile coverage-gap polygons and administrative boundaries, and applies Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to convert dispersed building points into shared and isolated terminal demand units. Six scenarios crossing two neighbourhood radii ($\epsilon = 100$ m, 200 m) and three minimum-cluster thresholds (MinPts = 2, 3, 4) generate a planning envelope rather than a single point estimate. Applied to Thailand, the framework identifies 199,762 building footprints across approximately 65,891 km² of coverage-gap area in 59 provinces, with estimated serviceable subscribers ranging from 19,118 to 52,746 units. Tak, Chiang Mai and Mae Hong Son consistently rank as the three highest-demand provinces across all scenarios. The framework provides a transparent, replicable evidence base for market sizing, deployment prioritisation and Universal Service Obligation planning, and is transferable to other countries with comparable coverage-gap conditions.

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1. Introduction

Thailand has built one of the most developed broadband infrastructures in Southeast Asia. Mobile-cellular networks cover more than 99% of the population, individual internet penetration is approximately 90% (ITU, 2026), and the country records mobile and fixed broadband speeds that are sufficient to support digital banking, e-government, online education, remote work and entertainment for the majority of the population. These aggregate indicators position Thailand among the leading broadband economies in the region (Ookla, 2026a).

Beneath these favourable national statistics, however, a persistent set of geographic coverage gaps remains. Remote locations such as mountainous areas, deep forests, national parks, protected zones, border districts and small offshore islands continue to lie outside the practical reach of terrestrial mobile networks (GSMA, 2026). The persistence of these coverage white spots is arguably not a question of radio technology alone. It is more likely a problem of network economics: the cost of towers, backhaul, energy supply, land access and recurrent maintenance is high relative to the limited revenue expected from a small and geographically dispersed potential user base (GSMA, 2016; GSMA Intelligence, 2025).

The commercial emergence of Low Earth orbit (LEO) satellite broadband since 2019 has introduced a new connectivity option that is specifically suited to such locations. Modern LEO services provide high throughput, low latency comparable to terrestrial fixed broadband and increasingly affordable user terminals, making satellite broadband a credible primary connectivity choice where terrestrial expansion is technically or economically unfeasible (McMahon et al., 2025; Ookla, 2026a; Starlink, 2025). LEO performance now matches a wide range of everyday digital applications.

A complementary development is the maturation of large-scale geospatial datasets derived from satellite imagery using deep learning. Such datasets identify the precise location of buildings in urban, rural and remote areas, including locations where conventional records or address registers are incomplete (Google Research, 2021; Sirko et al., 2023). When overlaid with terrestrial mobile coverage maps, building footprints can be used to isolate the specific structures that fall outside terrestrial coverage and that may therefore represent candidate satellite broadband demand. Geospatial intelligence thereby moves the analysis from broad population statistics to building-by-building demand identification.

A naive count of every building inside the coverage-gap layer would, however, overstate true subscriber demand. In Thai rural settlements, a single household commonly comprises several buildings within one compound, such as a main residence, a kitchen, livestock sheds, agricultural storage and auxiliary structures. Counting every footprint as

an independent subscriber would conflate buildings with subscribers and overstate the addressable demand. To address this, the present study applies Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (Schubert et al., 2017) to group nearby buildings into potential subscriber units, so that one cluster of proximate structures is counted as a single shared satellite terminal demand unit, while isolated buildings are counted as individual demand units.

The objective of this paper is to develop and apply an empirical geospatial framework for identifying and quantifying serviceable satellite broadband subscribers in remote areas of Thailand. Thailand is selected as the case study because it exemplifies a country with high overall broadband achievement that nevertheless contains residual remote-area gaps suitable for satellite-based service. This study accordingly addresses the following question: how can multi-source geospatial big data, combined with density-based spatial clustering, be used to identify and quantify serviceable satellite broadband subscribers in remote areas with no terrestrial mobile coverage in Thailand?

The remainder of the paper is organised as follows. Section 2 reviews relevant literature on Thailand's geographic and broadband context, terrestrial mobile coverage mapping, the emergence of LEO satellite broadband, building-footprint datasets and DBSCAN clustering. Section 3 describes the methodology, including data layers, processing steps and scenario design. Section 4 presents the empirical results, including the six DBSCAN scenarios, the leading provinces by estimated serviceable subscribers and a discussion of implications. Section 5 concludes this paper.

2. Literature Review

2.1 Thailand's geographic context

Thailand is located in mainland Southeast Asia, sharing borders with Myanmar to the north and west, the Lao People's Democratic Republic to the northeast, Cambodia to the southeast and Malaysia to the south. The country covers approximately 513,120 km² and is administratively organised into 77 provinces (changwat) and 928 districts (amphoe), spread across four major geographic regions (HDX, 2025; Royal Thai Government, 2024).

The Northern region is dominated by mountain ranges that form the southern extension of the Himalayan foothills. The Northeastern region (Isan) is characterised by a plateau, with low hills, agricultural land and an extensive border. The Central region is dominated by the Chao Phraya river basin, the country's most productive agricultural and economic zone. The Southern region is a narrow peninsula bordered by the Andaman Sea to the west and the Gulf of Thailand to the east and contains numerous offshore islands. This regional diversity has direct implications for broadband planning because

mountainous, forested and island areas reduce the feasibility of terrestrial network expansion (OECD, 2023).

2.2 Broadband status in Thailand

Thailand exhibits one of the highest aggregate broadband achievements in Southeast Asia. According to the International Telecommunication Union, mobile-cellular networks cover more than 99% of the population, 4G mobile broadband coverage reaches a comparable share of the population, and individual internet penetration is approximately 90% (ITU, 2026).

Subscription indicators reinforce this picture. According to the National Broadcasting and Telecommunications Commission (NBTC), Thailand had approximately 115 million mobile subscriptions and over 10 million fixed broadband subscriptions in 2024 (NBTC, 2024).

Network performance is equally competitive. According to the Ookla Speedtest Global Index, in Q4 of 2025, Thailand recorded a median fixed broadband download/upload speed of roughly 281/236 Mbps and a median mobile download/upload speed of roughly 136/15 Mbps, placing the country among the leading economies in Asia–Pacific for fixed broadband performance (Ookla, 2026a).

2.3 Mobile coverage mapping, white spots and the economics of rural connectivity

While national indicators are strong, spatially explicit coverage maps reveal that residual unserved locations are concentrated in specific geographies. Mobile coverage maps produced by network operators, industry bodies and international organisations provide a supply-side reference for identifying these so-called white spots (GSMA, 2026; ITU, 2025). The GSMA Network Coverage Maps (GSMA, 2026) are widely used reference sources for visualising mobile network availability across countries. Figure 1 illustrates the terrestrial mobile coverage map of Thailand used in this study, highlighting the residual white-spot mask.



Figure 1. Mobile terrestrial coverage map of Thailand (GSMA, 2026)
(white spots indicate mobile terrestrial coverage gap).

For the purposes of this study, in Figure 1, a white spot is operationally defined as a terrestrial location where the coverage-gap layer indicates outside reported terrestrial mobile broadband coverage. Sea areas are excluded from the analysis because maritime connectivity involves different user types, mobility patterns and regulatory considerations. Islands are retained when they form part of Thailand's terrestrial administrative territory and contain buildings in the coverage-gap layer. Based on the layers used in this study, approximately 12.84% of Thailand's terrestrial area lies within the residual mobile coverage gap, predominantly in mountainous, forested, border and remote island zones.

Persistent geographic coverage gaps remain. Mountainous areas, deep forests, national parks, border districts and small offshore islands continue to lie outside the practical reach of terrestrial mobile networks (GSMA, 2026). These white spots persist largely because rural deployment economics are unfavourable: low population density, limited disposable income, costly long-distance backhaul across difficult terrain and weak supporting infrastructure make build-out economically prohibitive relative to the small, expected user base (GSMA, 2016; GSMA Intelligence, 2025). Specifically, the revenue opportunity for new base stations in these rural or remote areas can be up to 10 times lower than that of an equivalent urban site. Compounding this lower revenue potential are significantly higher expenses; the capital costs (CapEx) for rural deployment can be up to 2 times higher (1.25x to 2.0x), while operating costs (OpEx) can be up to 3 times higher (1.5x to 3.0x) compared to urban areas (GSMA, 2016).

2.4 The emergence of LEO satellite broadband

The deployment of large LEO satellite constellations since 2019 has transformed the economics of satellite broadband. Starlink, operated by SpaceX, has emerged as the dominant LEO broadband provider, with more than 10,000 active satellites in orbit and more than 9 million subscribers worldwide as of 2025 (Johan, 2026; Marek, 2025; McMahon et al., 2025; Pultarova et al., 2026; Sheetz, 2024; Starlink, 2025). Competing LEO constellations, including Eutelsat OneWeb and Amazon Leo (Project Kuiper), are at different stages of deployment and are expected to broaden the competitive landscape for satellite broadband services in the coming years (OECD, 2024; Eutelsat, 2026; Amazon Staff, 2026).

Terminal design has also evolved. The Starlink residential and Mini kits use compact phased-array antennas that can be installed by an end user, weigh between approximately 1 and 5 kilograms depending on the model and operate from a standard household power supply. The smallest and most affordable option, the Starlink Mini, introduced in 2024, integrates Wi-Fi and a phased-array antenna in a portable form factor and is now available for under USD 500 for new customers in the United States, with service plans starting from approximately USD 50 per month (Sheetz, 2024; Mashable, 2026).

Performance is now competitive with terrestrial fixed broadband. Starlink delivered a median download speed of approximately 104 Mbps, a median upload speed of approximately 14 Mbps and a median latency of approximately 45 milliseconds in the United States as of 2025 (Ookla, 2026b). Combined with global coverage, rapid installation and minimal civil works requirements, these performance metrics support a wide range of consumer, enterprise, maritime, aviation, defence and disaster-recovery use cases.

2.5 Geospatial demand proxies: Google Open Buildings

Estimating serviceable subscribers in remote areas requires a demand proxy that is more spatially precise than census or province-level population data. Building footprints derived from high-resolution satellite imagery using AI provide such a proxy. The Google Open Buildings dataset (Google Research, 2021; Sirko et al., 2023) identifies building polygons across large geographic areas, including regions where conventional cadastral records or address registers are incomplete, and is therefore particularly valuable for remote-area analysis in forested, mountainous and sparsely settled areas.



Figure 2. Google Open Buildings dataset visualization:
(a) satellite image; (b) detected building footprints.

The Google Open Buildings dataset provides only building polygons and confidence scores, without information on building type, address or other semantic details (Google Research, 2021). It therefore cannot differentiate residential buildings from agricultural, commercial, religious or auxiliary structures, nor can it confirm whether a building is occupied or in active use, since AI-derived footprint datasets generally contain limited semantic information (Oostwegel et al., 2025). In addition, structures located on or near water, such as floating houses or stilt platforms, may be falsely detected or removed during quality filtering. Because the Open Buildings dataset is derived from AI-based image detection, omission and commission errors may occur, particularly in mountainous, forested, or sparsely populated tropical environments. Some structures may be missed due to vegetation cover, while non-residential or auxiliary structures may be incorrectly interpreted as potential household locations. Accordingly, this study treats building footprints as proxy indicators rather than confirmed households and addresses these limitations through clustering, scenario-based interpretation and sensitivity analysis.

2.6 DBSCAN for converting building points into subscriber units

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) (Ester et al., 1996) is a spatial clustering algorithm that groups points based on local density. DBSCAN does not require the number of clusters to be specified in advance and is robust to outliers, which it explicitly labels as noise. The algorithm requires two parameters: ϵ (the maximum distance between two points to be considered neighbours) and MinPts (the minimum number of points required within an ϵ -neighbourhood for a point to qualify as a core point, from which a cluster can be expanded).

The principal advantages of DBSCAN are its ability to identify clusters of arbitrary shape, its tolerance to noise and its avoidance of an a priori cluster count. Its principal limitations are sensitivity to parameter choice and difficulty in handling clusters of widely varying density (Schubert et al., 2017). For the present application, these strengths are well aligned with the characteristics of rural settlement patterns, where clusters of buildings vary in shape and size and isolated structures are common.

In this study, each qualifying DBSCAN cluster of building footprints is interpreted as one shared satellite terminal demand unit, on the planning assumption that buildings located within a given neighbourhood can plausibly share access to a single satellite terminal via local distribution (for example, local short-range distribution infrastructure). Each DBSCAN noise point is interpreted as one isolated subscriber unit. The two critical parameters are ϵ (denoted EPS in the scenario notation below), the maximum neighbour distance in meters, and MinPts, the minimum number of buildings required to form a cluster. These parameters function as planning assumptions that jointly determine when shared access to a single satellite terminal is treated as plausible.

3. Methodology

The methodology consists of four steps: (i) extraction of terrestrial coverage-gap polygons, (ii) overlay with Google Open Buildings footprints to identify candidate buildings, (iii) DBSCAN clustering to convert building points into serviceable subscriber units requiring a satellite user terminal, and (iv) scenario-based sensitivity analysis using six combinations of ϵ and MinPts. The workflow is designed to be transparent and replicable so that every estimated subscriber unit can be traced to a building point, a coverage-gap location, a DBSCAN label and an administrative assignment. Spatial preprocessing and clustering analysis were conducted using the DBSCAN method. All spatial layers were projected into a meter-based coordinate reference system (UTM Zone 47N, EPSG:32647) prior to distance-based clustering analysis, so that the ϵ parameter expressed in metres corresponds to true planar distance.

3.1 *Extracting terrestrial coverage-gap polygons*

The first step extracts mobile terrestrial coverage-gap polygons. The terrestrial area of Thailand is intersected with operator and industry mobile-coverage layers to produce a residual polygon layer that represents locations where terrestrial mobile broadband coverage is not reported. This layer is treated as the universe of locations where satellite broadband may be the primary connectivity option. Sea areas are excluded from the study area in order to focus exclusively on terrestrial serviceable subscribers. Islands that form

part of Thailand's terrestrial administrative territory are retained when they contain buildings within the coverage-gap layer. Although a coverage map cannot guarantee actual signal quality at every location, it provides the most consistent supply-side reference for spatial planning at national scale, as shown in Figure 3, which illustrates the extracted mobile terrestrial coverage-gap polygons.

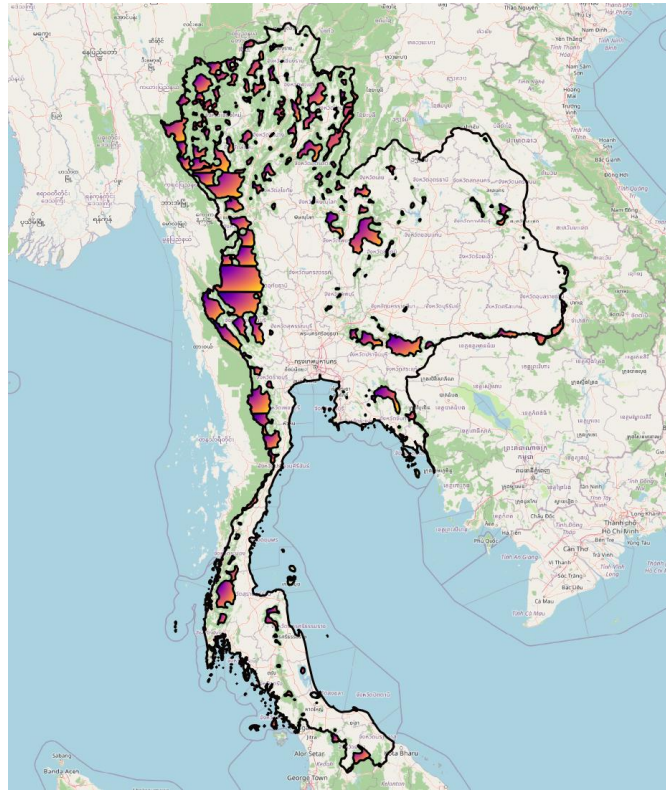


Figure 3. Mobile terrestrial coverage-gap polygons of Thailand (purple-pink gradient areas indicate mobile coverage gap polygons).

3.2 Overlaying coverage gaps with building footprints

The terrestrial coverage-gap polygons are overlaid with the Google Open Buildings dataset to identify building footprints located inside white-spot areas. Each retained building polygon is converted to a representative point (its centroid) before clustering, because DBSCAN operates on points and because point geometry simplifies neighbourhood distance calculations. Building points inside the coverage-gap mask are retained as candidate serviceable subscriber points, while points outside the mask are discarded. The candidate points are then spatially joined to provincial and district administrative boundaries, which allows subsequent results to be aggregated and reported at sub-national levels. A known limitation of this step is that Open Buildings does not reliably detect structures on open water; this is consistent with the terrestrial scope of the

framework but means that vessels and offshore platforms are not represented. The extracted building footprints overlaid with coverage-gap polygons are presented in Figure 4.

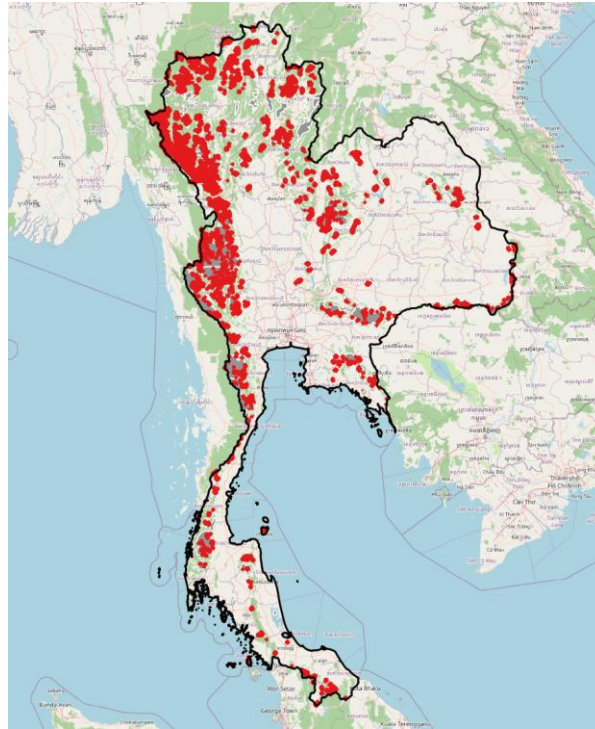


Figure 4. Building footprints overlaid with mobile terrestrial coverage-gap polygons (red spots indicate detected building locations).

3.3 DBSCAN clustering and subscriber conversion

Candidate building points are clustered using DBSCAN. For each scenario, one qualifying cluster is counted as a single shared satellite terminal demand unit, on the planning assumption that buildings located within a small neighbourhood can plausibly share one satellite terminal through local short-range distribution infrastructure. Each DBSCAN noise point is counted as a single isolated subscriber unit. The total estimated number of serviceable subscribers under any given scenario is therefore the sum of the number of qualifying clusters and the number of noise points:

$$\text{Total serviceable subscribers} = \text{Number of clusters} + \text{Number of noise points.}$$

Outputs are aggregated at the provincial level, and individual cluster and noise points are retained with their geographic coordinates (latitude/longitude) so that results can later be examined at the village or community level for deployment planning and field validation.

3.4 Scenario design and parameter interpretation

To explore the sensitivity of the serviceable subscriber demand estimate to the clustering assumptions, six scenarios are constructed by crossing two values of the neighbourhood radius ϵ (100 metres and 200 metres) with three values of the minimum-cluster-size threshold MinPts (2, 3 and 4). Each scenario is labelled by its parameter pair — for example, EPS100/MinPts2 denotes $\epsilon = 100$ m and MinPts = 2.

As described in section 2.6, the neighbourhood radius defines the maximum distance within which two buildings are assumed to share a single satellite terminal through local Wi-Fi or short-range wireless distribution. A radius of 100 m represents a conservative local-sharing planning assumption for proximate buildings; 200 m represents a more permissive spatial-sharing assumption for nearby rural settlement structures. The ϵ parameter is intended as a reproducible spatial planning heuristic rather than an engineering estimate of achievable wireless coverage. Values beyond 200 m are excluded because shared access over longer distances would require dedicated wireless infrastructure rather than commodity user equipment. Table 1 summarises the interpretation and planning meaning of each parameter value across the six scenarios.

Table 1. Interpretation of DBSCAN scenario parameters.

Scenario parameter	Value	Interpretation	Planning meaning
EPS	100 m	Shorter DBSCAN neighbourhood radius	Conservative local-sharing assumption
EPS	200 m	Wider DBSCAN neighbourhood radius	More permissive community-sharing assumption
MinPts	2	Core point requires 2 buildings within ϵ	Small groups can be treated as shared-terminal candidates
MinPts	3	Core point requires 3 buildings within ϵ	Moderate shared-access threshold
MinPts	4	Core point requires 4 buildings within ϵ	Conservative threshold; more buildings become isolated subscriber units

The MinPts threshold sets the minimum number of buildings that must lie within ϵ of a given building for that building to qualify as a core point and seed a shared-terminal cluster. Buildings that fall within ϵ of a core point but do not themselves meet the MinPts threshold are treated as border points and assigned to the same cluster, while buildings that are neither core nor border points are labelled as noise and interpreted as isolated subscriber units. At MinPts = 2, even a pair of neighbouring buildings qualifies as a shared unit. Raising the threshold to 3 requires a small-settlement-sized group, and MinPts = 4 demands a denser village-scale cluster before shared access is assumed. As MinPts increases, more buildings are reclassified as isolated subscriber units, reflecting a progressively conservative stance on the operational feasibility of terminal sharing among very small groups.

Together, the six scenarios span a sensitivity range from permissive sharing (EPS200/MinPts2) to conservative individual demand (EPS100/MinPts4) and are intended to be reported side by side rather than reduced to a single point estimate. Importantly, ϵ is a spatial planning assumption, not an engineering guarantee: it does not certify Wi-Fi range, line-of-sight clearance, or willingness to share, but provides an explicit, reproducible rule for converting building points into demand units. Table 1 summarises the interpretation and planning meaning of each parameter value, and Figure 5 and Figure 6 show an illustrative DBSCAN output distinguishing shared-terminal clusters from isolated noise points.

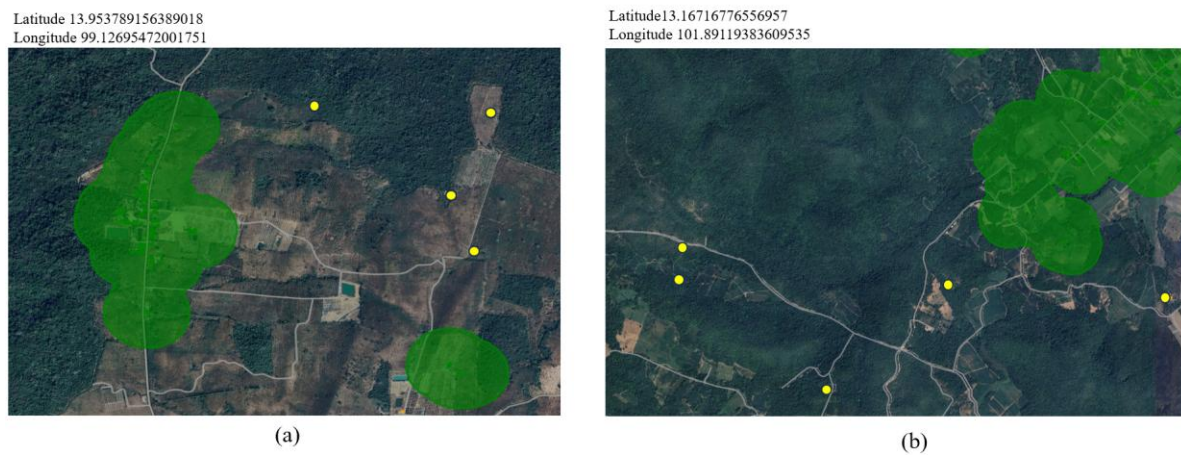


Figure 5. Illustrative DBSCAN output showing shared-terminal clusters (green areas) and isolated subscriber points (yellow spots).

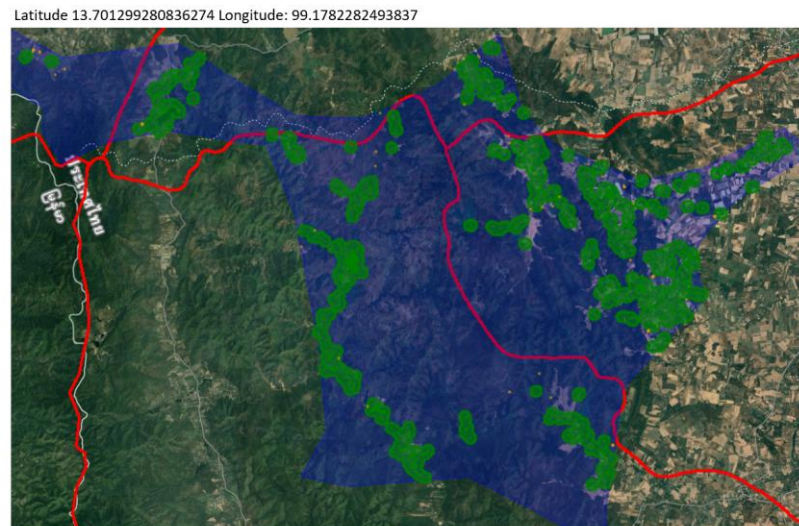


Figure 6. Illustrative DBSCAN output showing shared-terminal clusters (green areas) and isolated subscriber points (yellow spots) in the mobile coverage gap (blue areas).

4. Results and Implications

Intersecting the Google Open Buildings dataset with the mobile terrestrial coverage-gap polygons yields 199,762 building footprints in approximately 65,891 km² of coverage-gap area across 59 of Thailand's 77 provinces. This set of buildings forms the common candidate-point universe across all six DBSCAN scenarios; differences between scenario totals therefore arise purely from the choice of ϵ and MinPts, not from changes in the underlying study area.

4.1 Estimated serviceable subscribers across six DBSCAN scenarios

Table 2 reports the estimated number of serviceable satellite broadband subscribers under each of the six DBSCAN scenarios. The reported totals comprise two components: shared subscribers, defined as the number of qualifying clusters, and isolated subscribers, defined as the number of DBSCAN noise points.

Across the six scenarios, the estimated number of serviceable subscribers ranges from 19,118 units (EPS200/MinPts2) to 52,746 units (EPS100/MinPts4). Two consistent patterns are observed. First, holding MinPts constant, increasing ϵ from 100 m to 200 m reduces the total estimate, because a wider neighbourhood allows more buildings to be aggregated into shared clusters and thereby fewer buildings to be counted as individual subscribers. Second, holding ϵ constant, increasing MinPts from 2 to 4 increases the total estimate, because a stricter cluster threshold reclassifies more buildings as isolated noise points rather than members of shared clusters. The share of isolated subscribers in the total ranges from 68.29% (EPS200/MinPts2) to 92.71% (EPS100/MinPts4), indicating that buildings in Thailand's residual coverage gaps are predominantly spatially dispersed, even under the most permissive sharing assumption.

Table 2. Estimated serviceable satellite broadband subscribers across six DBSCAN scenarios.

Scenario	EPS (m)	MinPts	Total serviceable subscribers	Shared serviceable subscribers	Individual serviceable subscribers	(%) Shared serviceable subscribers	(%) Individual serviceable subscribers
EPS100/MinPts2	100	2	36,010	9,585	26,425	26.62%	73.38%
EPS200/MinPts2	200	2	19,118	6,063	13,055	31.71%	68.29%
EPS100/MinPts3	100	3	46,060	5,427	40,633	11.78%	88.22%
EPS200/MinPts3	200	3	29,296	4,566	24,730	15.59%	84.41%
EPS100/MinPts4	100	4	52,746	3,845	48,901	7.29%	92.71%
EPS200/MinPts4	200	4	37,068	3,548	33,520	9.57%	90.43%

4.2 Top provinces under illustrative scenarios

Because the six DBSCAN scenarios produce national totals that span a wide range — from 19,118 subscribers under the most permissive parameterisation (EPS200/MinPts2) to 52,746 under the most restrictive one (EPS100/MinPts4) — the EPS100/MinPts2 scenario (36,010 subscribers) is selected as the illustrative mid-range case for sub-national ranking. Table 3 accordingly reports the ten provinces with the highest estimated serviceable subscribers under this scenario. The table also reports the number of building footprints inside the coverage-gap layer, the decomposition between shared and isolated subscribers, and the total coverage-gap area for each province.

Under this scenario, Tak records the highest estimated serviceable subscribers (8,683 units), followed by Chiang Mai (6,391 units), Mae Hong Son (4,184 units), Kanchanaburi (2,433 units) and Nan (2,031 units). These provinces are not necessarily the most populous provinces in Thailand. Rather, they share three structural characteristics: extensive mountainous or forested terrain, the presence of national parks or protected areas and proximity to international borders. The presence of dispersed settlements and ethnic-minority communities in these areas further contributes to the high count of residual demand. Notably, Kanchanaburi has the second largest coverage-gap area among the top ten but ranks fourth in estimated demand, indicating that coverage-gap area alone is not a sufficient proxy for demand without considering the actual spatial distribution of buildings.

Table 3. Top ten provinces by estimated serviceable subscribers under the EPS100/MinPts2 scenario.

Rank	Province	Area (km ²)/ (%) of nationwide mobile coverage gap	Buildings in mobile coverage gap	Total serviceable subscribers	Shared serviceable subscribers/ (%)	Individual serviceable subscribers/ (%)
1	Tak	9,250.9/ (14.04%)	51,132	8,683	1,993/(22.95%)	6,690/(77.05%)
2	Chiang Mai	5,504.1/ (8.35%)	36,708	6,391	1,442/(22.56%)	4,949/(77.44%)
3	Mae Hong Son	5,797.1/ (8.80%)	31,618	4,184	1,005/(24.02%)	3,179/(75.98%)
4	Kanchanaburi	8,413.5 / (12.77%)	18,745	2,433	971/(39.91%)	1,462/(60.09%)
5	Nan	3,262.9/ (4.95%)	8,458	2,031	457/(22.50%)	1,574/(77.50%)
6	Prachuap Khiri Khan	1,668.2/ (2.53%)	5,465	1,232	473/(38.39%)	759/(61.61%)
7	Loei	678.9/ (1.03%)	4,827	1,151	276/(23.98%)	875/(76.02%)
8	Chiang Rai	862.9/ (1.31%)	6,686	1,093	269/(24.61%)	824/(75.39%)
9	Phayao	1,069.7/ (1.62%)	2,400	996	249/(25.00%)	747/(75.00%)
10	Phrae	1,317.7/ (2.00%)	2,639	754	199/(26.39%)	555/(73.61%)

As shown in Table 3, the top five provinces alone account for approximately 65.87% of total estimated demand (23,722 out of 36,010 units), indicating a high geographic concentration of unserved buildings. All ten leading provinces are characterised by mountainous terrain or extensive forest cover, which explains both the persistent terrestrial coverage gaps and the dispersed settlement patterns that drive demand. Notably, the share of isolated subscriber units mostly exceeds 70% for those provinces, confirming that buildings within these coverage gaps are predominantly scattered rather than clustered. This pattern suggests that deployment strategies in these areas will need to balance shared-terminal approaches for the few denser settlements with individual-terminal provision for the majority of isolated households.

Table 4 reports the top three provinces under each of the six DBSCAN scenarios. The ranking is striking in its stability: Tak, Chiang Mai and Mae Hong Son consistently occupy the first, second and third positions across all six scenarios. While the absolute number of estimated subscriber units changes with EPS and MinPts, the relative ordering does not.

Table 4. Top three provinces by estimated serviceable subscribers across six DBSCAN scenarios.

Scenario	1st (Province/ total serviceable subscribers)	2nd (Province/ total serviceable subscribers)	3rd (Province/ total serviceable subscribers)
EPS100/MinPts2	Tak / 8,683	Chiang Mai / 6,391	Mae Hong Son / 4,184
EPS200/MinPts2	Tak / 4,509	Chiang Mai / 3,136	Mae Hong Son / 2,233
EPS100/MinPts3	Tak / 10,912	Chiang Mai / 8,042	Mae Hong Son / 5,183
EPS200/MinPts3	Tak / 7,094	Chiang Mai / 4,848	Mae Hong Son / 3,356
EPS100/MinPts4	Tak / 12,132	Chiang Mai / 8,938	Mae Hong Son / 5,695
EPS200/MinPts4	Tak / 9,003	Chiang Mai / 6,165	Mae Hong Son / 4,268

From Table 4, the consistency of this ranking across all six scenarios carries two key observations. First, the top-three ordering is entirely insensitive to the choice of ϵ and MinPts, which indicates that the concentration of unserved buildings in Tak, Chiang Mai and Mae Hong Son — all located in northern Thailand, where mountain ranges and extensive forest cover limit terrestrial network deployment — is a structural feature of the country's geography rather than an artefact of a particular parameter choice. Tak's total ranges from 4,509 units under the most permissive clustering assumption (EPS200/MinPts2) to 12,132 units under the strictest (EPS100/MinPts4), a factor of roughly 2.7. Nevertheless, the proportional gap between these three provinces and the remaining provinces persists across all scenarios. Together, these observations suggest that the three provinces constitute a robust tier of highest residual demand for satellite

broadband, providing a degree of confidence in prioritising them for further analysis regardless of which specific DBSCAN configuration is ultimately adopted.

4.3 Implications from the result

The framework supports several practical applications. From a market-assessment perspective, the planning envelope of 19,118 to 52,746 estimated terminal demand units defines the order of magnitude of the addressable LEO satellite broadband market in Thailand's residual coverage gaps. Service providers can use the scenario range to compare individual-terminal, shared-terminal and mixed deployment models, and to prioritise the provinces in which initial market entry is most likely to be commercially viable.

From a policy perspective, the geographically resolved outputs of the framework can directly support Universal Service Obligation (USO) planning. Because the framework retains latitude and longitude information at the cluster and noise-point level, candidate locations can be aggregated to district, sub-district and even village levels. This enables regulators to identify specific communities in which targeted public intervention — such as subsidised installation, shared-terminal community access points or co-funded deployment with operators — may close residual coverage gaps more cost-effectively than further terrestrial expansion. The framework does not by itself determine commercial viability or affordability, but provides an evidence base on which subsequent feasibility studies, field measurements and policy decisions can be built.

The findings also suggest a sequencing strategy for deployment and policy. Provinces that rank consistently high across all six scenarios, notably Tak, Chiang Mai and Mae Hong Son, represent natural starting points for both commercial pilots and USO interventions using satellite connectivity. Provinces that appear in the top ten under the illustrative scenario but with significantly varying estimates across scenarios may benefit from preliminary field surveys to refine the choice of EPS and MinPts before larger-scale rollout.

The framework estimates spatially potential demand units rather than actual broadband subscribers or realized adoption. The results do not incorporate affordability, willingness to pay, electricity availability, terminal visibility constraints, or operator commercial strategy. In addition, the DBSCAN-derived clusters have not yet been validated against household surveys or field observations. Accordingly, the framework should be interpreted as a planning-oriented geospatial demand proxy rather than a direct forecast of subscriber adoption.

5. Conclusion

This paper has developed an empirical geospatial framework for estimating serviceable satellite broadband subscribers in terrestrial coverage gaps, with remote areas in Thailand as the case study. The framework integrates building-footprint data from Google Open Buildings, terrestrial mobile coverage-gap polygons and administrative boundaries, and applies DBSCAN clustering to convert dispersed building points into shared and isolated subscriber units. By construction, it makes its planning assumptions explicit and avoids reducing the analysis to a single point estimate.

Applied to Thailand, the framework identifies 199,762 building footprints across approximately 65,891 km² of terrestrial coverage-gap area in 59 provinces, and produces a planning envelope of 19,118 to 52,746 estimated serviceable subscribers across six DBSCAN scenarios. Tak, Chiang Mai and Mae Hong Son consistently rank as the three highest-demand provinces, indicating that the broad spatial pattern of residual demand is robust to scenario assumptions.

The contribution of this study is both empirical and methodological. Empirically, it transforms aggregate national broadband indicators into a spatially explicit, building-level view of residual demand and identifies the specific provinces in which satellite broadband can have the greatest impact. Methodologically, it offers a transparent and replicable geospatial workflow that can be updated as new building footprints, coverage maps, administrative boundaries or field-validation data become available. The framework can be transferred to other countries with comparable coverage-gap conditions and is therefore well suited to inform regional studies of digital inclusion.

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